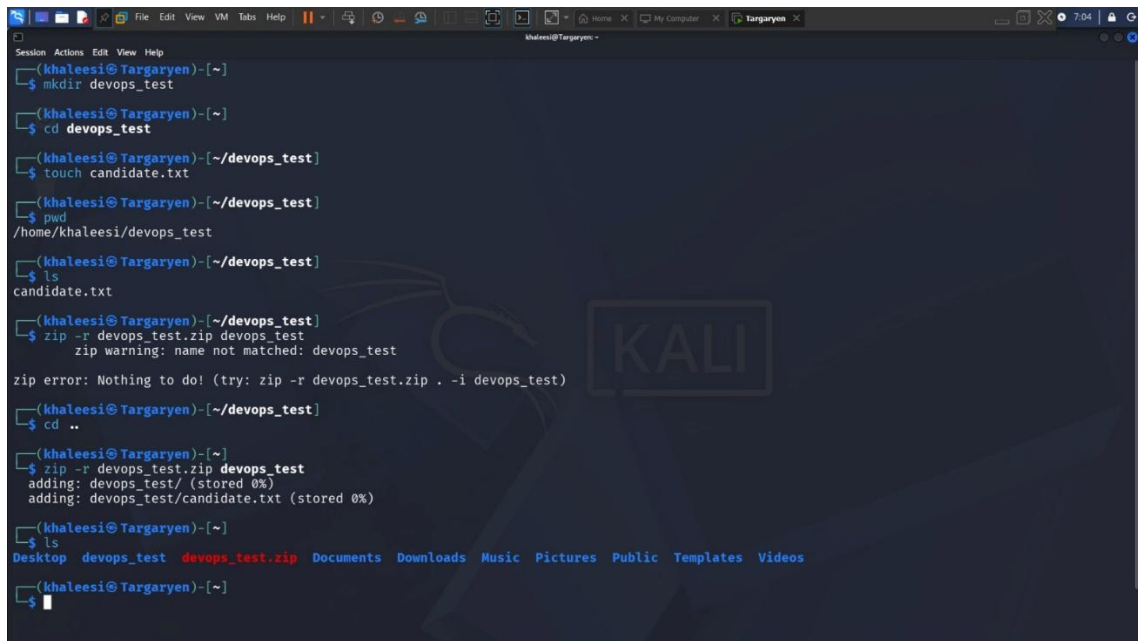


TASK 1:

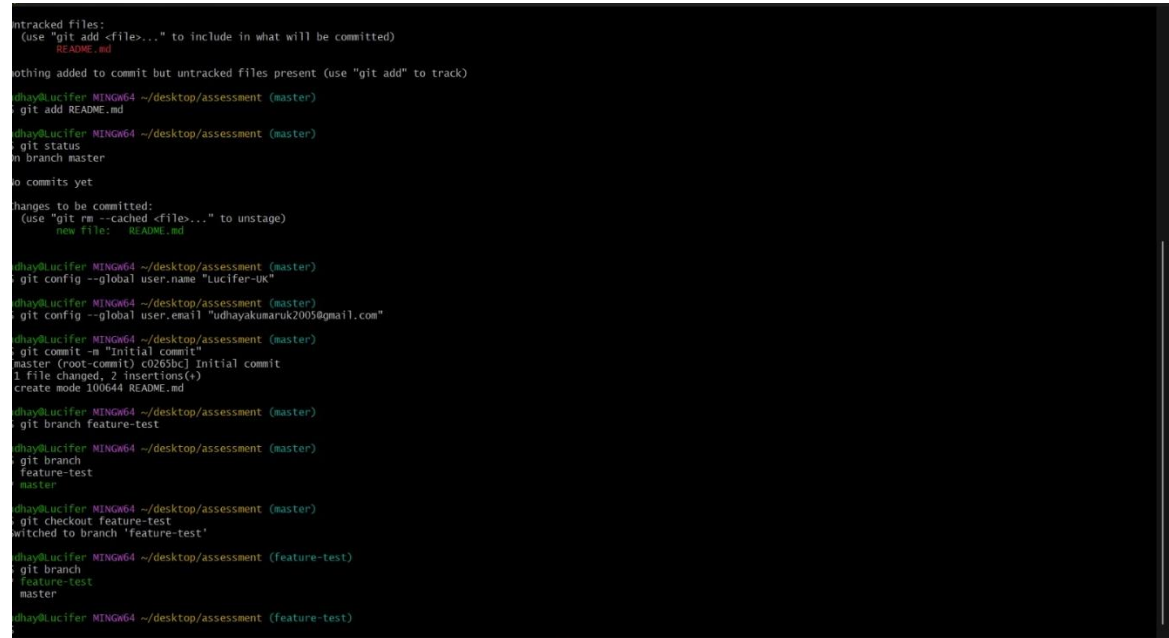
LINUX BASICS



```
khaleesi@Targaryen:~$ mkdir devops_test
khaleesi@Targaryen:~$ cd devops_test
khaleesi@Targaryen:~/devops_test$ touch candidate.txt
khaleesi@Targaryen:~/devops_test$ pwd
/home/khaleesi/devops_test
khaleesi@Targaryen:~/devops_test$ ls
candidate.txt
khaleesi@Targaryen:~/devops_test$ zip -r devops_test.zip devops_test
zip warning: name not matched: devops_test
zip error: Nothing to do! (try: zip -r devops_test.zip . -i devops_test)
khaleesi@Targaryen:~/devops_test$ cd ..
khaleesi@Targaryen:~$ zip -r devops_test.zip devops_test
adding: devops_test/ (stored 0%)
adding: devops_test/candidate.txt (stored 0%)
khaleesi@Targaryen:~$ ls
Desktop  devops_test  devops_test.zip  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

TASK 2:

Git Fundamentals



```
untracked files:
  (use "git add <file>..." to include in what will be committed)
  README.md

nothing added to commit but untracked files present (use "git add" to track)
dhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git add README.md

dhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git status
On branch master

no commits yet

changes to be committed:
  (use "git rm --cached <file>..." to unstage)
  new file:   README.md

dhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git config --global user.name "Lucifer-UK"

dhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git config --global user.email "udhayakumaruk2005@gmail.com"

dhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git commit -m "Initial commit"
[master (root-commit) c0265bc] Initial commit
1 file changed, 2 insertions(+)
create mode 100644 README.md

dhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git branch feature-test

dhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git branch
  feature-test
* master

dhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git checkout feature-test
Switched to branch 'feature-test'

dhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$ git branch
  feature-test
* master

dhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
```

```

MINGW64~/c/Users/udhay/desktop/assessment
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  README.md

nothing added to commit but untracked files present (use "git add" to track)
udhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git add README.md
$ git status
On branch master
no commits yet
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   README.md

udhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git config --global user.name "Lucifer-UK"
udhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git config --global user.email "udhayakumaruk2005@gmail.com"
udhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git commit -m "Initial commit"
[master (root-commit) c0265bc] Initial commit
 1 file changed, 2 insertions(+)
 create mode 100644 README.md
$ git branch
$ git branch feature-test
$ git branch
* feature-test
  master
udhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git checkout feature-test
Switched to branch 'feature-test'
udhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$ git branch
* feature-test
  master
udhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$

```

```

MINGW64~/c/Users/udhay/desktop/assessment
$ git branch
  feature-test
* master
udhay@Lucifer MINGW64 ~/desktop/assessment (master)
$ git checkout feature-test
Switched to branch 'feature-test'
udhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$ git branch
* feature-test
  master
udhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$ git status
On branch feature-test
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
udhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$ git add README.md
udhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$ git commit -m "Updated README in feature-test branch"
[feature-test 9c2feed] Updated README in feature-test branch
 1 file changed, 4 insertions(+)
udhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$ git log
commit 9c2feed34a6baa7f3e4e5ea290acc643db5b2b51 (HEAD -> feature-test)
Author: Lucifer-UK <udhayakumaruk2005@gmail.com>
Date:   Mon Jun 29 18:32:17 2026 +0530

    Updated README in feature-test branch

commit c0265bc81b74910beddd59a59289ada5b161455a (master)
Author: Lucifer-UK <udhayakumaruk2005@gmail.com>
Date:   Mon Jun 29 18:29:26 2026 +0530

    Initial commit
udhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$ git log --oneline
9c2feed (HEAD -> feature-test) updated README in feature-test branch
c0265bc (master) Initial commit
udhay@Lucifer MINGW64 ~/desktop/assessment (feature-test)
$

```

TASK 3:

DOCKER PRACTICAL

```
C:\WINDOWS\system32\cmd. x + v
C:\Users\udhay\Desktop\dockertest>dir
Volume in drive C is OS
Volume Serial Number is 25C0-25E0

Directory of C:\Users\udhay\Desktop\dockertest

29-06-2026 18:04 <DIR> .
29-06-2026 18:03 <DIR> ..
29-06-2026 18:04      143 index.html
                1 File(s)      143 bytes
                2 Dir(s)  62,423,248,896 bytes free

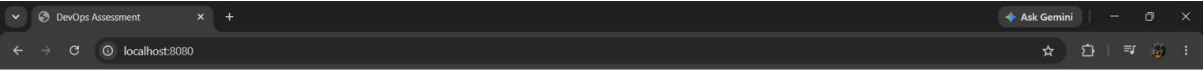
C:\Users\udhay\Desktop\dockertest>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
e95a6c7ea7d4: Pull complete
1645c1e06f46: Pull complete
df68ee7e7a00: Pull complete
1b30016634d5: Pull complete
acf093e7a04f: Pull complete
cd9307c9ecd8: Pull complete
fcb6fd84b2a0: Pull complete
1cf7d051b485: Download complete
e2c07e54e55a: Download complete
Digest: sha256:ec4ed8b5299e5e90694af7750eb6dff2627317d30544d056b0371f8082f7bce
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

What's next:
View a summary of image vulnerabilities and recommendations → docker scout quickview nginx

C:\Users\udhay\Desktop\dockertest>docker run -d -p 8080:80 -v %cd%:/usr/share/nginx/html --name devops-nginx nginx
f2a7b69cf4274b5e777eb89cc42c4fde9caebae3786a04f70cdf0dda4e5b9a

C:\Users\udhay\Desktop\dockertest>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
f2a7b69cf427   nginx     "/docker-entrypoint..." 14 seconds ago Up 14 seconds  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp devops-nginx

C:\Users\udhay\Desktop\dockertest>
```



DevOps Intern Assessment

TASK 4:

Troubleshooting & AWS Concepts

Question 1

Scenario:

You can't access a deployed application.

3 Possible Causes

1. The application service isn't running (maybe the server or container has stopped).
2. There's a network issue (like a wrong port, or a firewall/security group blocking traffic).
3. There's a deployment or configuration error (this could be due to incorrect environment variables or application setup).

First 3 Checks

1. Check if the application/service is up and running.

Example: Use commands like `systemctl status`, `docker ps`, or look for the application process.

2. Look at the application logs for any errors.

Example: You can check with `docker logs <container-id>` or look at the application log files.

3. Review the network settings.

Make sure the right port is open and that firewall/security group rules are allowing access.

Question 2

EC2

Amazon EC2 (Elastic Compute Cloud) provides virtual servers in the cloud for running applications. It allows users to choose from various instance types based on their computing needs.

S3

Amazon S3 (Simple Storage Service) is an object storage service that's designed to securely store and retrieve files like images, videos, backups, and documents.

IAM

AWS IAM (Identity and Access Management) is used to securely manage users, groups, roles, and permissions for accessing AWS resources.

Question 3

A Docker container stops right after it starts.

Two Possible Reasons

1. The main application inside the container crashes due to an error or incorrect setup.
2. The main process finishes right away, which causes the container to exit.

How to Investigate

Check the container logs with:

```
docker logs <container_id>
```

This will help you identify any application errors.

Inspect the container details using:

```
docker inspect <container_id>
```

Also, take a moment to double-check the Dockerfile, startup command, and environment variables to ensure everything is set up correctly.